Computer Networks

Lecture 2: The Physical Layer

Demo Lecturer

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Recap: Where We Are

- Five layers: application, transport, network, link, **physical**
- The physical layer moves raw **bits** across a medium
- Today: how fast can a link carry bits?

Signals and Bandwidth

- A signal varies over time; its content spans a range of frequencies
- **Bandwidth** B = the width of the usable frequency range, in hertz (Hz)
- A channel passes frequencies from f_1 to f_2, so $B = f_2 - f_1$

Example: a telephone line passing 300 Hz to 3300 Hz has $B = 3000$ Hz.

Bandwidth, Data Rate, Latency

Quantity	Unit	Meaning
Bandwidth B	Hz	width of the frequency range
Data rate	bits/s	how many bits per second we send
Latency	s	time for one bit to get across

Careful: in everyday speech "bandwidth" is often used for the data rate. In this subject they are different quantities.

Nyquist: A Noiseless Channel

For a noiseless channel of bandwidth B Hz and signals with V distinct levels:

$$\text{max data rate} = 2B \log_2 V \text{ bits/s}$$

Example. $B = 3000$ Hz, binary signal ($V = 2$):

$$2 \times 3000 \times \log_2 2 = 6000 \text{ bits/s}$$

More signal levels give more bits per symbol – but noise limits how many levels we can tell apart.

Shannon: A Noisy Channel

With signal-to-noise ratio S/N (a plain ratio, not decibels):

$$C = B \log_2\left(1 + \frac{S}{N}\right) \text{ bits/s}$$

Example. $B = 3000$ Hz and 30 dB: $S/N = 10^{30/10} = 1000$, so

$$C = 3000 \times \log_2(1001) \approx 29\,900 \text{ bits/s}$$

No coding scheme can beat C, however many signal levels are used.

Transmission Media

Medium	Typical use	Main limit
Twisted pair	office LAN, last mile	attenuation over distance
Optical fibre	backbone links	cost of installation
Radio	Wi-Fi, mobile	shared spectrum, interference

Summary

- Bandwidth (B, Hz) and data rate (bits/s) are different quantities
- Nyquist: $2B \log_2 V$ for a noiseless channel
- Shannon: $B \log_2(1 + S/N)$ for a noisy channel – an upper limit
- Next: the link layer and how a shared medium is accessed